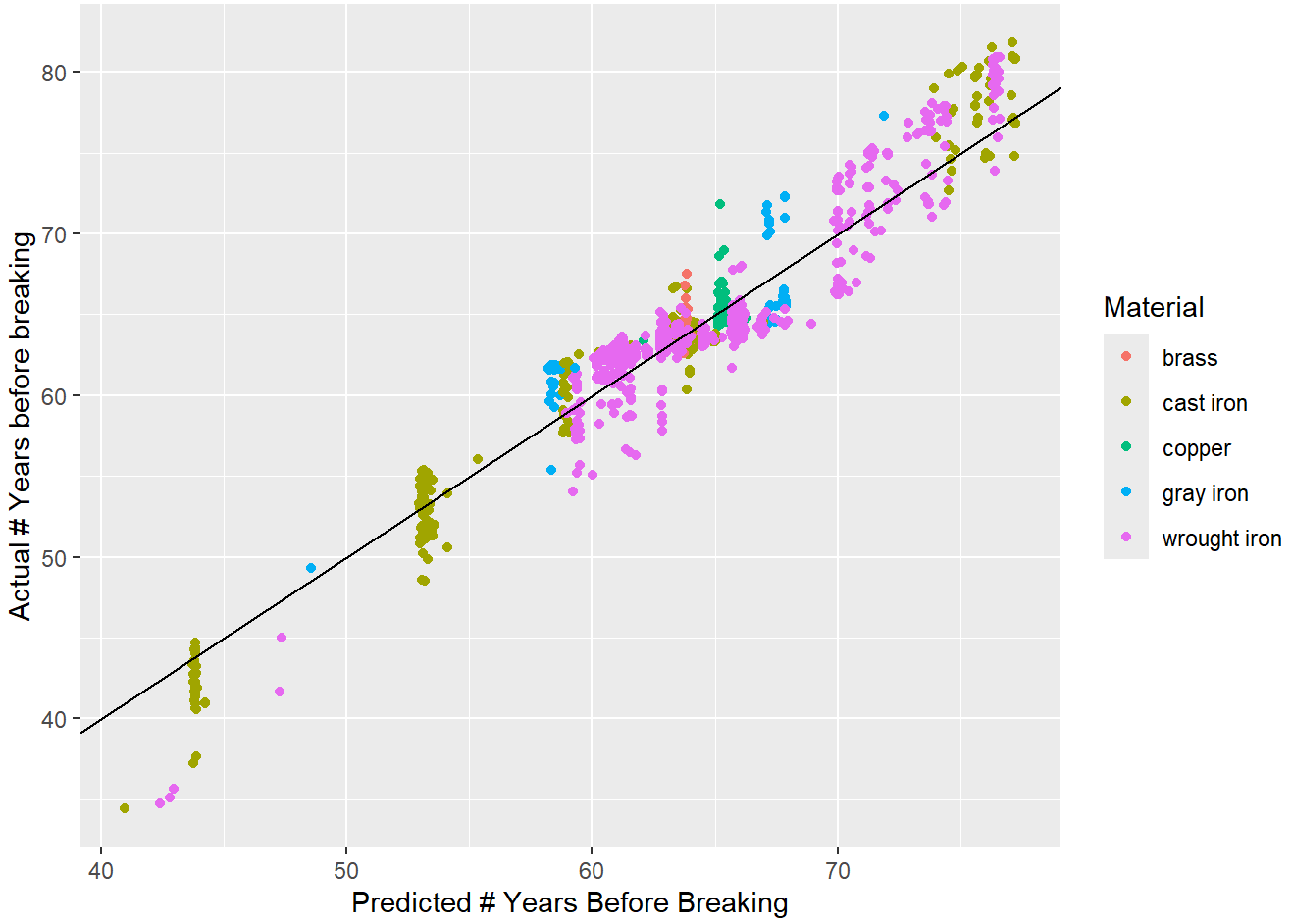
Visualization

```
fit3 <- lm(Years~Surface+Material+Length+`Lay date`, data=data[set==1,])
pred_v_s3 <- predict(fit3, newdata = val)
MSPE_v_s3 <- mean((as.numeric(val[,12])- pred_v_s3)^2)
ggplot(data=val) +geom_point(mapping=aes(x=pred_v_s3, y=as.numeric(Years),
        colour=Material))+ geom_abline(a=0,b=1)+
labs(x="Predicted # Years Before Breaking", y="Actual # Years before breaking")
```

```
Warning in geom_abline(a = 0, b = 1): Ignoring unknown parameters: `a` and `b`
```




```
data2 <- read_csv("STAT440Project1_files/Train_update3.csv")
```

```
Warning: One or more parsing issues, call `problems()` on your data frame for details,
e.g.:
  dat <- vroom(...)
  problems(dat)

Rows: 43039 Columns: 12
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
```

```
chr  (3): Pipe ID, Material, Surface
dbl  (7): GPS x1, GPS y1, GPS x2, GPS y2, Leaked, Age, Length
date (2): Lay date, Currdate

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

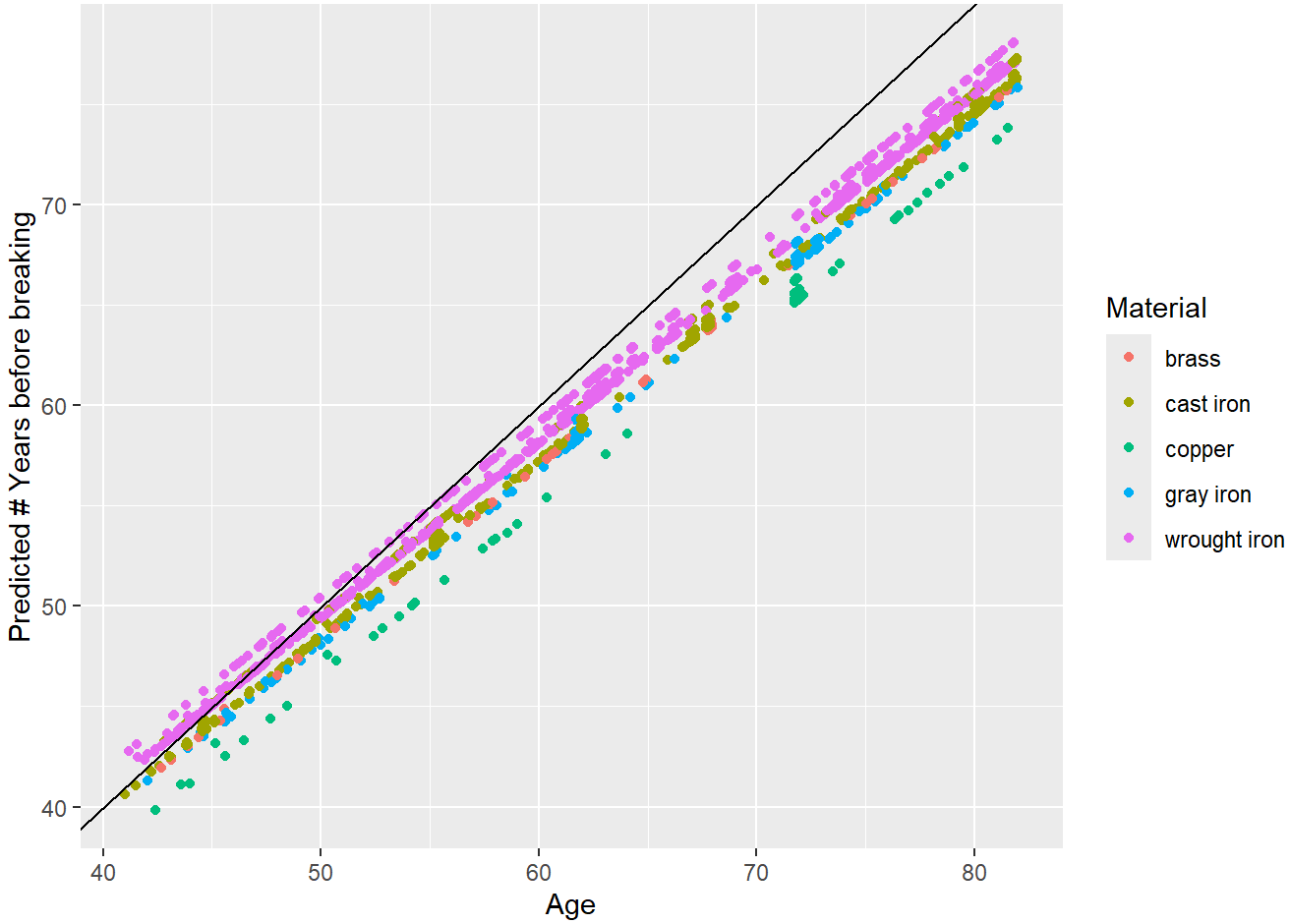
```
na.omit(data2)
```

```
# A tibble: 43,016 × 12
   `Pipe ID` `Lay date` Material  Surface   `GPS x1` `GPS y1` `GPS x2` `GPS y2`
   <chr>     <date>     <chr>     <chr>        <dbl>    <dbl>    <dbl>    <dbl>
 1 P01001    1955-03-11 copper    grassland    6136.    5021.    6124.    5021.
 2 P01002    1971-09-19 cast iron water        6512.    4148.    6537.    4202.
 3 P01003    1971-09-18 cast iron water        6537.    4202.    6518.    4202.
 4 P01004    1982-05-22 cast iron structure    2837.    3378.    2849.    3405.
 5 P01005    1971-07-03 cast iron water        5117.    4273.    5126.    4297.
 6 P01006    1971-10-05 cast iron structure    3740.    5080.    3728.    5120.
 7 P01007    1971-10-06 cast iron structure    3734.    5099.    3722.    5099.
 8 P01008    1971-10-05 cast iron structure    3734.    5099.    3728.    5077.
 9 P01009    1982-05-29 cast iron water        2570.    4299.    2615.    4299.
10 P01010    1971-07-20 cast iron water        5772.    4257.    5760.    4283.
# ℹ 43,006 more rows
# ℹ 4 more variables: Leaked <dbl>, Currdate <date>, Age <dbl>, Length <dbl>
```

```
#data that havent leaked and arent polyurethane
data2nl <- filter(data2,Leaked=="0"& Material!="polyurethane")
pred <- predict(fit3, newdata = data2nl)
ggplot(data=data2nl) +geom_point(mapping=aes(x=Age, y=pred,
      colour=Material))+ geom_abline(a=0,b=1)+labs(x=
        "Age", y="Predicted # Years before breaking")
```

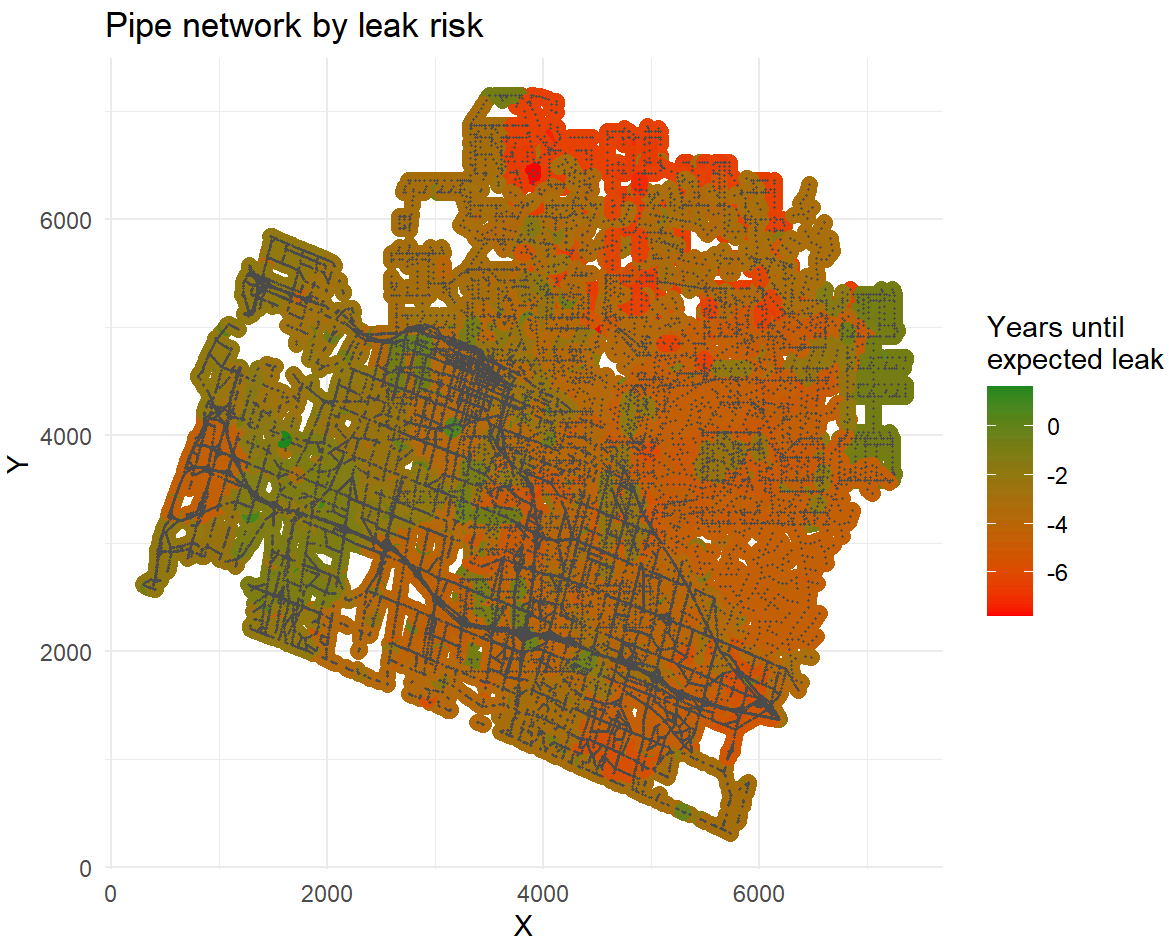
```
Warning in geom_abline(a = 0, b = 1): Ignoring unknown parameters: `a` and `b`

Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_point()`).
```



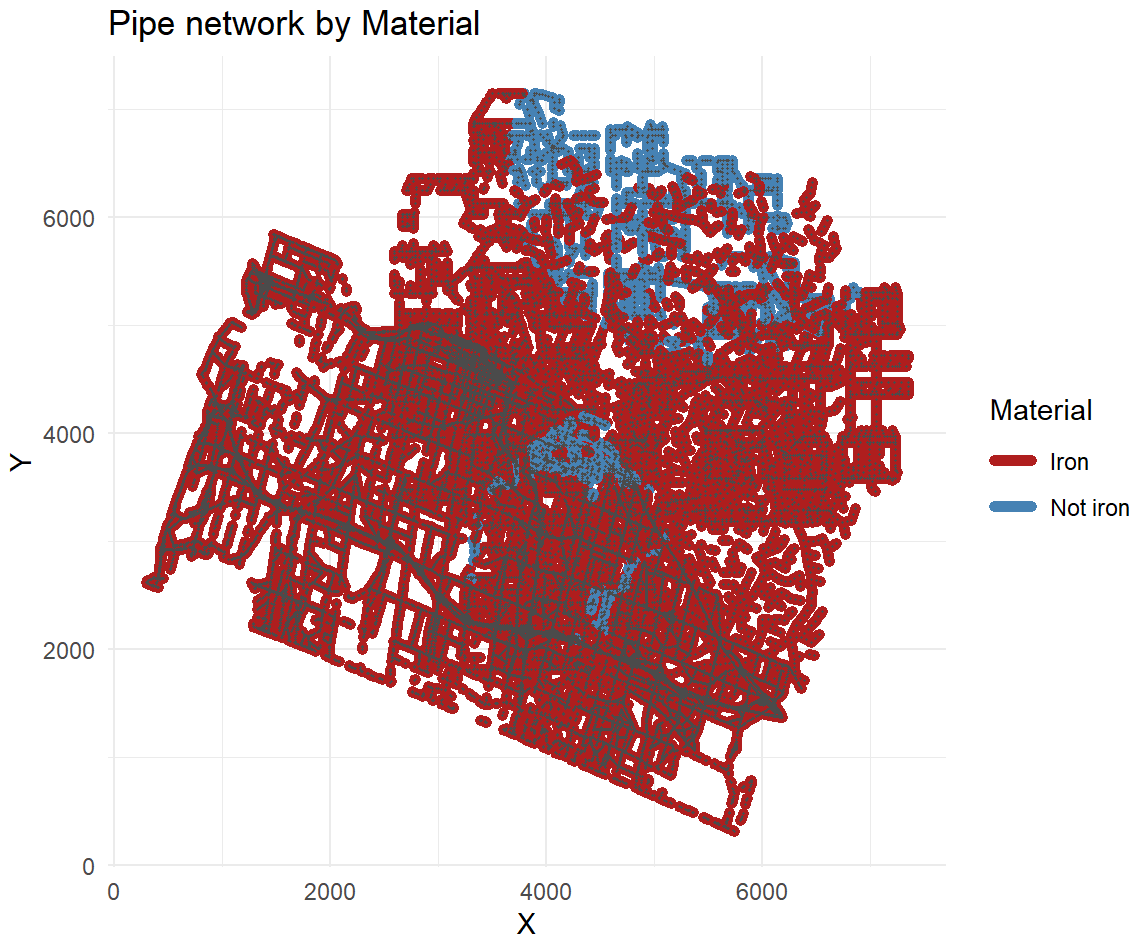

```
data2nl <-cbind(data2nl,(pred-data2nl$Age))
colnames(data2nl)[13]<-"yrs2leak"
```

```
ggplot(data2nl) +
  geom_segment(
    aes(x = `GPS x1`, y = `GPS y1`, xend = `GPS x2`, yend = `GPS y2`,
        color = yrs2leak),
    linewidth = 3, lineend = "round"
  ) +
  geom_point(aes(x = `GPS x1`, y = `GPS y1`), color = "grey30", size = 0.2) +
  geom_point(aes(x = `GPS x2`, y = `GPS y2`), color = "grey30", size = 0.2) +
  scale_color_gradient(
    low = "red", high = "forestgreen",
    name = "Years until\nexpected leak"
  ) +
  coord_equal() +
  labs(title = "Pipe network by leak risk", x = "X", y = "Y") +
  theme_minimal()
```




```
data2nl$is_iron <- ifelse(grepl(" iron",data2nl$Material), "Iron", "Not iron")

ggplot(data2nl) +
  geom_segment(
    aes(x = `GPS x1`, y = `GPS y1`, xend = `GPS x2`, yend = `GPS y2`,
        color = is_iron),
    linewidth = 2, lineend = "round"
  ) +
  geom_point(aes(x = `GPS x1`, y = `GPS y1`), color = "grey30", size = 0.2) +
  geom_point(aes(x = `GPS x2`, y = `GPS y2`), color = "grey30", size = 0.2) +
  scale_color_manual( values = c("Iron" = "firebrick", "Not iron" = "steelblue"),
    name = "Material") +
  coord_equal() +
  labs(title = "Pipe network by Material", x = "X", y = "Y") +
  theme_minimal()
```


Pipe network by Material
6000
4000
2000
0
Y
0
2000
4000
6000
X
Material
Iron
Not iron